

Ruiyi Yang

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Profile

I am a PhD candidate in Computer Science at the University of New South Wales (UNSW), supervised by Prof. Flora Salim, A/Prof. Hao Xue, and A/Prof. Imran Razzak. My research focuses on knowledge graphs, large language models, retrieval-augmented generation, and multi-agent systems, with interests in complex reasoning, self-evolving agents, spatio-temporal intelligence, and trustworthy agentic AI.

Education

Ph.D.	University of New South Wales , Sydney, Australia Computer Science Supervisors: Prof. Flora Salim, A/Prof. Hao Xue, A/Prof. Imran Razzak	Apr. 2024 – Nov. 2027
M.S.	University of New South Wales , Sydney, Australia Data Science and Decisions	Sep. 2022 – Dec. 2023
B.Eng.	Beihang University , Beijing, China Software Engineering	Sep. 2017 – June. 2022

Work Experience

University of New South Wales , Research Officer Advisors: Prof. Salil Kanhere, Dr. Rahat Masood, Prof. Flora Salim	Sydney, Australia Jul. 2025 – Present
- Worked in the project <i>GuardianAgent: A Trustworthy Personal Agentic AI Platform</i>, focusing on privacy-preserving multi-agent system design. - Designed the privacy protection framework, including agent coordination, user data handling, and trustworthy interaction mechanisms. - Coordinated user study design, system revision, and manuscript preparation; one first-author paper was accepted to Findings of EMNLP 2026 and another is under submission.	
University of New South Wales , Casual Academic	Sydney, Australia Jul. 2024 – Present
- Taught tutorials for multiple undergraduate and postgraduate computer science and data science courses. - Mentored honours and master's students, co-supervising "Taste of Research" projects with Prof. Flora Salim.	
Cresstec Pty Ltd , Software Engineer Intern	Sydney, Australia Dec. 2023 – Present
- Developed and maintained software modules for HDAC temperature control systems. - Built and improved React-based dashboards and data visualisation components for energy monitoring, scheduling, and operational analytics for both internal and customer usage.	
Jina AI , Software Engineer Intern	Beijing, China Sept. 2021 – June. 2022
- Maintained and fixed executors compatible with the core Jina framework. - Developed a cross-modal search example supporting both image and text indexing, enabling users to search images with text queries or retrieve descriptions from uploaded images. - Contributed to DocsQA, a question-answering system built on company product documentation. - Participated in the CLIP-as-Service project, integrating models into the framework to support bilingual retrieval and query understanding.	

Publications

- **Ruiyi Yang**, Zechen Li, Hao Xue, Imran Razzak, Flora D. Salim. *MAGE: Multi-Agent Self-Evolution with Co-Evolutionary Knowledge Graphs*. Advances in Neural Information Processing Systems (NeurIPS), 2026. Poster.
- **Ruiyi Yang**, Hao Xue, Imran Razzak, Shirui Pan, Hakim Acid, Flora D. Salim. *Divide by Question, Conquer by Agent: SPLIT-RAG with Question-Driven Graph Partitioning*. EMNLP 2026 (Main Conference).
- **Ruiyi Yang**, Gayathri Lihinikaduarachchi, Rahat Masood, Flora D. Salim, Salil S. Kanhere. *GuardianAgent: Policy-Conditioned Risk-Adaptive Anonymization with Verified Adversarial Escalation*. Findings of EMNLP 2026.
- **Ruiyi Yang**, Hao Xue, Imran Razzak, Hakim Acid, Flora D. Salim. *RELOOP: Recursive Retrieval with Multi-Hop Reasoner and Planners for Heterogeneous QA*. Findings of EMNLP 2026.
- **Ruiyi Yang**, Hao Xue, Imran Razzak, Flora D. Salim. *Beyond Single Pass, Looping Through Time: KG-IRAG with Iterative Knowledge Retrieval*. Proceedings of the ACM Web Conference (WWW), 2026. Oral Presentation.
- **Ruiyi Yang**, Lihuan Li, Hao Xue, Flora D. Salim. *STAR: Failure-Aware Markovian Routing for Multi-Agent Spatiotemporal Reasoning*. Preprint, 2026.
- Lihuan Li, Wilson Wongso, Baiyu Chen, Hao Xue, **Ruiyi Yang**, Yifan Duan, Xiachong Lin, Yang Song, Flora Salim. *TrajPrism: A Multi-Task Benchmark for Language-Grounded Urban Trajectory Understanding*. Preprint, 2026.
- **Ruiyi Yang**, Flora D. Salim, Hao Xue. *SSTKG: Simple Spatio-Temporal Knowledge Graph for Interpretable and Versatile Dynamic Information Embedding*. Proceedings of the ACM Web Conference (WWW), 2024. Oral Presentation.

Honors & Awards

- ACM Web Conference 2026 Oral Presentation 2026
- UNSW Higher Degree by Research Scholarship 2024 – Present
- ACM Web Conference 2024 Oral Presentation 2024
- Master Degree with Excellence, University of New South Wales 2023

Skills

- Languages:** Chinese(native); English(proficient); French(1.5 year full time learning experience); Japanese(JLPT-N2)
- Programming:** Python, C#, JavaScript, Java, SQL, Bash, HTML/CSS
- AI & Machine Learning:** LLM, RAG, Multi-Agent Systems, Self-Evolving Agents, Knowledge Graphs, Time-Series Analysis
- Software Development:** React.js, Flask, Vue.js, Backend-Frontend Integration
- Data & Databases:** PostgreSQL, MySQL, Neo4j, Cypher, Pandas, NumPy